

Case 1: From Distributed Proof to Multi-Part Problem

Step 1: Raw Document Input

4.3. Main result

Theorem 1. Let $p \in [1, \infty)$. If $s > d_p^*(\mu)$, then $\mathbb{E}[W_p] \lesssim n^{-1/s}$. If $t < d_*(\mu)$, then $W_p \gtrsim n^{-1/t}$.
[... narrative transition ...]

4.4. Proof of upper bound

Proposition 4. [Statement...]
Proof. [Proof of Prop 4...]

Proposition 5. [Statement...]
Proof. [Proof of Prop 5...]

Corollary 1 (Thm 1, upper bound). [Statement...]
Proof. [Proof of Cor 1...]

4.5. Proof of lower bound

Proposition 6. [Statement...]
Proof. [Proof of Prop 6...]

Corollary 2 (Thm 1, lower bound). [Statement...]
Proof. [Proof of Cor 2...]

Step 2: Dependency Extraction & Marker Insertion

Target: Proof of Theorem 1

Extracted Dependencies:

Prop. 4 Prop. 5 Cor. 1 Prop. 6 Cor. 2

----- Sanitized Macro-Proof(Regex Output) -----

4.4. Proof of upper bound

The upper bound of Theorem 1 follows from Proposition 5, below.
[...] All the upper bounds we prove rely on the following fundamental estimate [...]

<MARKER: PROP_4_ENCAPSULATED>

The key step in proving the upper bound of Theorem 1 is giving a bound for $\mathbb{E}[W_p^p(\mu, \hat{\mu}_n)]$ in terms of the quantity [...]

<MARKER: PROP_5_ENCAPSULATED>

<MARKER: COROLLARY_1_ENCAPSULATED>

4.5. Proof of lower bound

Our asymptotic lower bounds involving $d_p^*(\mu)$ follow from a much simpler argument. [...] The following proposition is adapted from [18] and forms the core of the lower bound.

<MARKER: PROP_5_ENCAPSULATED>

<MARKER: COROLLARY_1_ENCAPSULATED>

Step 3: QA Synthesis (Medium-Level Output)

Question: Multi-Part Problem

Part 1 (Prop 4): For any Bore set $\mathcal{S}$ and dyadic partition $\mathcal{Q}^k$, prove $\mathbb{E}[\sum_{Q_i^k} |\mu(Q_i^k) - \hat{\mu}_n(Q_i^k)|] \leq 2(1 - \mu(\mathcal{S})) + \sqrt{|\{i : Q_i^k \cap \mathcal{S} \neq \emptyset\}|/n}$.

Part 2 (Prop 5): Assume $p \in [1, \infty)$. Given $d_e(\mu, e^{sp/(s-2p)}) \leq s$ for $\varepsilon \leq \varepsilon'$, prove $\mathbb{E}[W_p^p(\mu, \hat{\mu}_n)] \leq C_1 n^{-p/s} + C_2 n^{-1/2}$.

Part 3 (Cor. 1 – Upper Bound):

Conclude that if $s > d_p^*(\mu)$, then $\mathbb{E}[W_p(\mu, \hat{\mu}_n)] \lesssim n^{-1/s}$.

Part 4 (Prop 6): Suppose $\exists \varepsilon', \tau, t > 0$ s.t. $\mathcal{N}_e(\mu, \tau) \geq \varepsilon^{-t}$. Prove $W_p^p(\mu, v) \geq \tau 4^{-p} n^{-p/t}$ for any $|\text{supp}(v)| \leq n$.

Part 5 (Cor. 2 – Lower Bound):

Conclude that if $t < d_*(\mu)$, then $W_p(\mu, v) \gtrsim n^{-1/t}$.

Reference Proof (Sketch):

- **Prop 4:** Binomial variance on $\mathcal{S}'$ + trivial bound on $X/\mathcal{S}'$; apply Cauchy-Schwarz.
- **Prop 5:** Construct T_k via covering numbers; sum bounds across scales k .
- **Cor. 1:** If $s > d_p^*(\mu)$, choose ε' s.t. $d_e \leq s$; apply Prop. 5 and take p -th root.
- **Prop. 6:** Choose $\varepsilon = n^{-1/t}/2$; mass outside balls around $\text{supp}(v)$ is $\geq \tau$.
- **Theorem 1:** Combine Cor. 1(upper) and Cor. 2(lower).
- **Cor. 2:** By def. of $d_e(\mu)$, obtain constants satisfying Prop. 6; conclude.

Extracted modules (Independent Objects)

Proposition 4 [Statement + Proof]

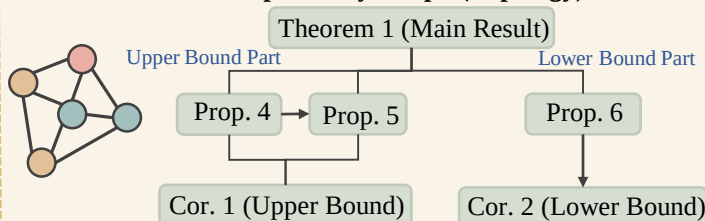
Proposition 5 [Statement + Proof]

Corollary 1 [Statement + Proof]

Proposition 6 [Statement + Proof]

Corollary 2 [Statement + Proof]

Dependency Graph (Topology)



Final Multi-Part Problem



- ✓ Integrates all dependencies into a coherent multi-part problem.
- ✓ Preserves logical structure of distributed proof.
- ✓ Evaluates both upper and lower bounds systematically.